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Inventor(s)	McCarthy; Martin Franklin et al.

Systems and methods for collecting and transporting biofuel components

Abstract

A method of capturing and disposing of fats, oil and/or grease (“FOG”) is disclosed. The method includes providing a vacuum truck to a user, wherein the vacuum truck has a tank with (1) a vacuum hose; (2) an inlet to the tank connected directly to the hose; (3) an outlet from the tank; (4) and one or more mixing components. The method also calls for positioning an absorbent material within the tank that is capable of absorbing a quantity of FOG. Next, FOG is vacuumed from a grease trap via the hose and mixed with the absorbent material to create a biofuel. Finally, the biofuel is discharged from the outlet of the tank into a containment device at a burn facility and subsequently burned. Alternatively, the absorbent material may be contained within a containment device at the burn facility and mixed with the FOG at the facility before being burned.

Inventors: McCarthy; Martin Franklin (Charlotte, NC), McCarthy; Sean Thomas (Charlotte, NC)

Applicant: McCarthy; Martin Franklin (Charlotte, NC); McCarthy; Sean Thomas (Charlotte, NC)

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Primary Examiner: Williams; Patrick C

Attorney, Agent or Firm: Shumaker, Loop & Kendrick, LLP

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to collecting and transporting a biofuel product having fat, oil and/or grease components. In its preferred forms, the collection and transportation of the product is accomplished through the use of a vacuum truck or other similar mobile containment device that may include various other components such as heating and/or mixing elements.

BACKGROUND

(2) The National Pretreatment Program implements Clean Water Act requirements to control pollutants that are introduced into publicly owned treatment works (“POTWs”). As part of this program, EPA has promulgated General Pretreatment Regulations that require the establishment of

State and local pretreatment programs to control pollutants, which pass through or interfere with POTW treatment processes or may contaminate POTW sewage sludge. Meeting these requirements may require elimination of interference caused by the discharge to POTWs of Fat, Oil, and Grease (FOG) from food service establishments (FSE). More specifically, the Pretreatment Program regulations at 40 CFR § 403.5 (b) (3) prohibit “solid or viscous pollutants in amounts which will cause obstruction” in the POTW and its collection system. EPA's Report to Congress on combined sewer overflows (CSOs) and sanitary sewer overflows (SSOs) identified that “grease from restaurants, homes, and industrial sources are the most common cause (47%) of reported blockages. Grease is problematic because it solidifies, reduces conveyance capacity, and blocks flow.”

(3) Controlling FOG discharges will help POTWs prevent blockages that impact CSOs and SSOs, which cause public health and water quality problems.

(4) FOG wastes are generated at food service establishments as byproducts from food preparation, and cleaning activities for pans, dishes, utensils and other surfaces. FOG captured on site is generally classified into two broad categories. The first type is yellow grease that is the byproduct of deep-frying, and often captured in large containers, then ultimately sold into the reuse market. The second type of FOG are the fat, oil and grease that are washed down the sink and floor drains into the Grease Trap. These fats, oils and grease are a result of cleaning pans, plates, utensils and other grease-laden surfaces in the food service establishment. The annual production of grease trap waste is massive. Currently the EPA estimates between 23,000 and 75,000 Sanitary Sewer Overflows per year. Food service establishments create volumes of FOG that run from 800 to 1,700 pounds per year. Furthermore, one source indicates that Americans produce 13 pounds of F.O.G. per capita per year.

(5) Food service establishments can adopt a variety of best management practices or install interceptor/collector devices to control and capture the FOG material before discharge to the POTW collection system. For example, instead of discharging yellow grease to POTWs, food service establishments often accumulate this material for pick up by consolidation service companies for re-sale or re-use in the manufacture of tallow, animal feed supplements, fuels, or other products.

(6) Additionally, food service establishments can install interceptor/collector devices (e.g., grease traps) in order to accumulate FOG on-site and prevent it from entering the POTW collection system. In many cases, an establishment that implements best management practices will realize financial benefit through a reduction in their required grease interceptor and trap maintenance frequency.

(7) Likewise, more and more POTWs are addressing FOG discharges by imposing mandatory measures of various types, including inspections, periodic grease pumping, stiff penalties, and even criminal citations for violators, along with ‘strong waste’ monthly surcharges added to restaurant sewer bills.

(8) As a separate matter, large quantities of motor vehicle oils and lubricants also end up in water supplies for various reasons. Motor vehicle oils and lubricants thus fall within the definition of fats, oils and/or grease as used in this application.

(9) Pretreatment programs are developing and using inspection checklists for both food service establishments and POTW pretreatment inspectors to control FOG discharges. Additionally, EPA identified typical numeric local limits controlling oil and grease in the range of 50 mg/L to 450 mg/L with 100 mg/L as the most commonly reported numeric pretreatment limit.

(10) With this information in mind, it is apparent that while there has been some progress in collecting and disposing of FOG, much more needs to be done. In particular, there is a need for a safe and effective method of transporting the biofuel product composed largely of FOG and similar natural, biodegradable materials having high energy density such that the FOG is usable as fuel in a wide variety of applications and at low cost.

SUMMARY

(11) Accordingly, the invention of this application employs a specially designed mobile vehicle, such as but not limited to a vacuum truck into which FOG can be collected, transported and discharged from the vehicle for use as a fuel product.

(12) In one example, an elongate tube or absorbent mat geotextile product is used to contain sphagnum peat or mushroom compost materials, and to maximize contact surface area with the FOG materials in, for example, a grease trap. The sphagnum peat, mushroom compost, and/or orange peels is obtained from select locations in the United States or Canada known for this type of specialized product. As used in this application, peat, mushroom and similar materials into which the FOG is absorbed are referred to generally and broadly as “capture materials” or absorbent materials. In certain aspects, it is contemplated that orange peels may be used as a capture or absorbent material. Likewise, the absorbent material may be a polymeric material such as a polyolefinic material and more preferably a polypropylene having oleophilic and hydrophobic properties that are ideal to absorb fat, oil, and/or grease.

(13) Several products suitable for use in the FOG tube described in this application are “Dry All” wood fiber, sphagnum peat moss processed and sold by Integrity Absorbent Products or shredded mushrooms. In particular, the peat moss product is an all organic hydrocarbon absorbent, manufactured from large fiber sphagnum peat moss. The manufacturing process produces a product which becomes both oleophilic, absorbing hydrocarbons and hydrophobic, i.e., repelling water. Due to its fibrous structure and processing, peat absorbs hydrocarbons quickly on contact by virtue of its wicking capillary action and encapsulates oil on contact. This makes peat ideal for hydrocarbon cleanup both on open water and land applications. Peat absorbs up to eight times its weight. This volume will vary based on the hydrocarbon being absorbed and the temperature. In certain aspects, orange peels have the same oleophilic and hydrophobic characteristics as those mentioned above.

(14) This type of natural cleansing and separation is one of the unique features of this invention and why it will be useful to restaurants, industrial facilities and car repair shops that struggle with the maintenance of grease traps and oil spills. Once trapped in the tube or mat, the product can be easily and compactly shipped to a location for disposal, incineration or further processing, including processing the materials for use as fuel.

(15) Sewage Sludge Incineration (SSI) is becoming a safe and effective alternative around densely populated municipalities where land application of sewage sludge is less desirable. One of the benefits of the sphagnum peat FOG, or loose fuel product, absorbent tubes and mats is that they comprise a high BTU fuel that can be used to increase the efficiency of SSI processes. In addition to providing a better and more efficient way for collecting and disposal of FOG, the product can separate the higher density grease and oil so that it can be disposed of in a landfill, and/or burned as fuel in a sludge incinerator or other furnace.

(16) Moreover, it is another object of the invention to provide a biofuel product having fat, oil and/or grease constituents.

(17) It is another object of the invention to provide a biofuel product having fat, oil and/or grease constituents that provide enhanced environmental remediation by avoiding the need to dispose of the materials in landfills or other geographical locations.

(18) It is another object of the invention to provide a product having fat, oil and/or grease constituents that provide for the ability to utilize natural, renewable, biodegradable materials to produce a fuel suitable for a wide variety of uses.

(19) It is another object of the invention to provide having fat, oil and/or grease constituents that burned as fuel in a sludge incinerator or in other facilities such as furnaces.

(20) It is another object of the invention to utilize containers, such as tubes, mats and other configurations to capture spills of oil and other hydrocarbons at vehicle repair facilities that can then be burned along with the oil or other hydrocarbons as fuel.

(21) These and other objects and advantages are achieved by providing a biofuel product having

constituents selected from the group consisting of fat, oil and/or grease components, a container formed of a biodegradable material having a multiplicity of openings of a size and shape adapted for allowing the fat, oil and/or grease components to pass through the openings to an interior area of the container, an absorbent capture material positioned in the container and holding a quantity of the fat, oil and/or grease, the container, capture material and fat, oil and/or grease collectively comprising the biofuel product.

(22) According to another aspect of the invention, the container is a biodegradable geotextile.

(23) According to another aspect of the invention, the container is constructed of a biodegradable yarn selected from the group consisting of cotton, hemp, ramie and jute.

(24) According to another aspect of the invention, the openings in the container have an apparent opening size (AOS) of 0.25 mm to 0.5 mm. In this aspect, the AOS of the container is smaller than the size(s) of the absorbent material positioned therein such that the absorbent material remains within the container until being released/removed from the container.

(25) According to another aspect of the invention, the container is a tube.

(26) According to another aspect of the invention, the container is a three-dimensional box-like mat.

(27) In certain aspects, the grease absorbent material is oleophilic and hydrophobic material/matrix configured to absorb fats, oils, and/or greases while repelling and/or filtering out water from, for example, a mixture of water, fats, oils, and greases. According to another aspect of the invention, the fat, oil and/or grease absorbent capture material is selected from the group consisting of an organic material such as sphagnum peat, mushroom compost, and orange peels. According to another aspect of the invention, the fat, oil and/or grease capture material is selected from an absorbent porous, polymeric material that is preferably oleophilic and hydrophobic. In this aspect, the absorbent porous, polymeric material is a polyolefinic material and more preferably a polypropylene having oleophilic and hydrophobic properties that are ideal to absorb fat, oil, and/or grease.

(28) According to another aspect of the invention, the fat, oil and/or grease is present in a range of between 88-75 percent and the capture material is present in a range of between 12 and 25 percent.

(29) According to another aspect of the invention, the fat, oil and/or grease and the absorbent capture material is processed according to a process selected from the group of processes consisting of compressing the fat, oil and/or grease and the absorbent capture material into pellets, logs, cakes, shredding and granulating.

(30) According to another aspect of the invention, the fat, oil and/or grease, the absorbent capture material and the container collectively comprise the biofuel product.

(31) According to another aspect of the invention, the biofuel product contains between 88-75 percent FOG and between 12 and 25 percent capture material.

(32) According to another aspect of the invention, the range of B.T.U. output of the biofuel product is 12,500 to 20,000 B.T.U. per pound.

(33) According to another aspect of the invention, a biofuel product is provided having constituents selected from the group consisting of fat, oil and grease components, and including a container formed of a biodegradable geotextile having a multiplicity of openings have an apparent opening size (AOS) of 0.25 mm to 0.5 mm and adapted for allowing the fat, oil and/or grease components to pass through the openings to an interior area of the container. An absorbent capture material is positioned in the container and holds a quantity of the fat, oil and/or grease, the container, capture material and fat, oil and/or grease collectively comprising the biofuel product. In certain aspects, the grease absorbent material positioned in the container is oleophilic and hydrophobic wherein the absorbent material will absorb fats, oils, and/or greases while concurrently repelling water. The fat, oil and/or grease absorbent capture material is selected from naturally occurring products such as sphagnum peat, mushroom compost, and orange peels. Alternatively, the fat, oil and/or grease capture material is selected from an absorbent porous, polymeric material. In this aspect, the

absorbent porous, polymeric material is a polyolefinic material and more preferably a polypropylene having oleophilic and hydrophobic properties that are ideal to absorb fat, oil, and/or grease. The fat, oil and/or grease present in a range of between 88-75 percent and the capture material present in a range of between 12 and 25 percent. The fat, oil and/or grease and the absorbent capture material can be presented in a multiplicity of forms including pellets, cakes, logs, or as shredded or granulated fuel.

(34) According to another aspect of the invention, the container includes a tether for positioning the container at an influent end of a source of fat, oil and/or grease during absorption of the fat, oil or grease into the capture material.

(35) According to another aspect of the invention, the fat, oil and/or grease is present in a range of between 88-75 percent and the capture material present in a range of between 12 and 25 percent, the fat, oil and/or grease and the absorbent capture material being in a multiple of forms including pellets, cakes, logs, or as shredded or granulated fuel and the range of B.T.U. output of the biofuel product is 12,500 to 20,000 B.T.U. per pound.

(36) According to another aspect of the invention, the container includes one closed end and an open end adapted for being closed after being filled with the capture material.

(37) In additional aspects, the FOG absorbent material may be used alone (i.e., excluding the previously mentioned container that contains the absorbent material and/or the absorbent material separated from the container) to absorb FOG when forming/producing the biofuel product (i.e., the absorbent material having FOG absorbed therein). When used in this manner, a user (e.g., technician) places a desired, predetermined amount of absorbent material within, for example, a grease trap (or other container having FOG therein). The user then allows an adequate time for the absorbent material to absorb and preferably become saturated with FOG thereby forming the biofuel product. Subsequently the user removes the biofuel product from, for example, the grease trap for subsequent use and/or processing steps to be used as the biofuel product. During the removal step of the biofuel product (i.e., absorbent material having FOG absorbed therein) in the above mentioned process, it is further envisioned that a positive pressure pump (e.g., a modified pool pump such as the Pentair line of pool pumps) and other removal aids will be used to further expedite the overall process to remove the biofuel from the grease trap in an expeditious and efficient manner.

(38) In the above-mentioned method, the technician arrives at the grease trap to be serviced and removes the grease trap's lid. (It is preferred that the grease trap is serviced at predetermined time intervals to maintain proper operability of the grease trap as well as to maintain manageable, efficient FOG removal during each servicing.) Next, the amount of absorbent material to be introduced into the grease trap will be determined by the technician visually inspecting and assessing the overall volume of the grease trap along with the ratio of FOG to brown water contained therein. The technician next introduces a sufficient amount of the loose absorbent material (e.g., 1 pound, 5 pounds, 10 pounds, 20 pounds, etc.—termed the “first amount” or “initial amount”) into the trap to allow for the desirable amount of absorption of FOG into the absorbent material. For example, each pound of absorbent material should absorb between four (4) pounds to nine (9) pounds of FOG per pound of absorbent material, and in preferred aspects, each pound of absorbent material should absorb between 7 pounds to 8 pounds of FOG per pound of absorbent material. In preferable aspects, the desirable amount absorption of FOG into the absorbent material is 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 98%, or 100% of the FOG within the grease trap.

(39) After a predetermined time period in which absorption of FOG into the absorbent material has occurred, the grease trap is again visually inspected and assessed to further determine whether any FOG remains, and if so, to further determine the ratio of FOG to brown water remaining in the grease trap. At that time, more absorbent material (a “second amount” of absorbent material) may be added to absorb the remaining FOG (i.e., FOG that was not absorbed by the previously provided

absorbent material). Subsequent amounts of absorbent material (e.g., “third amount,” “fourth amount,” “fifth amount,” etc.) may be added by the technician until the desirable amount of FOG absorption (and desired FOG to brown water ratio) has been reached. After reaching the desired FOG absorption amount in the absorbent material (and the desired FOG to brown water ratio), the absorbent material having FOG absorbed therein (biofuel) may be subsequently removed all at once by the removal step/process discussed further below.

(40) As an alternative to the above, an absorbing step followed by an immediate removal step may be used. In this aspect, the absorbent material (i.e., first amount or initial amount) is added to the grease trap and after a predetermined time period FOG is absorbed into the absorbent material forming the biofuel. Next, the biofuel (i.e., first amount of absorbent material having FOG absorbed therein) is removed from the grease trap. After removal of the absorbent material having FOG absorbed therein (i.e., biofuel) from the grease trap, the technician may again visually inspect and assess the grease trap to determine whether any additional FOG remains therein. If so, the technician may again, introduce the absorbent material (“second amount” of absorbent material) and allow for absorption of FOG into the second amount of absorbent material (biofuel). Subsequently, the second amount of absorbent material having FOG absorbed therein is removed from the grease trap. The above mentioned steps (i.e., absorbing FOG by the absorbent material followed by an immediate removing step until a desirable FOG to brown water ratio and/or FOG absorption amount in the absorbent material has been reached) may be repeated as desired by the technician.

(41) To further aid in the removing/removal steps (i.e., removing the biofuel comprising the absorbent material having FOG absorbed therein), additional removal tools and aids may be used. For example, the technician may use a rake or rake-like device to actuate the absorbent material, FOG, and/or brown water in the grease trap and to spread the absorbent material more evenly on the surface of the FOG/water mixture within the grease trap. This step advantageously increases the rate of absorption into the absorbent material and further expedites the FOG capture and removal process from the grease trap. During this step, the absorbent material having FOG absorbed therein is hydrophobic and will continue to float even when saturated with FOG.

(42) At this point, the technician may further utilize a positive pressure pump (a modified pool pump such as the Pentair line of pool pumps) to pump the biofuel (i.e., FOG laden absorbent material) from the top, or uppermost, portion of the grease trap and to remove the biofuel therefrom via a filter and/or physically remove (e.g., via a net or another similar removal device). During the removal process and while using the pump, the pump will beneficially circulate the brown water back into the grease trap thereby keeping the grease trap maximally efficient and effective. The maximum efficiency of the grease trap is achieved by keeping the grease trap full of water during the removal process.

(43) While pumping the FOG. laden absorbent material out of the grease trap, the technician moves the end of the hose around the top and/or uppermost sections of the grease trap surface until all of the biofuel (i.e., absorbent material having FOG absorbed therein) has been removed from the grease trap. The same outcome can be achieved with a net, filter/filtration, or similar device.

(44) After the biofuel (i.e., absorbent material laden with FOG and/or absorbent material having FOG absorbed therein) has been captured and removed from the grease trap and the unencumbered water drained from the positive pressure pump hoses into the grease trap, the grease trap lid is closed securely. Then the technician either transports the biofuel to another location or leaves the biofuel in a secure area for subsequent removal and processing.

(45) In certain aspects, the method of forming a biofuel by capturing fats, oil and/or grease in a grease trap are disclosed. In view of the above disclosures, this method may include comprising the steps of: (a) adding an absorbent material to the grease trap that has a mixture of water, fats, oil, and grease therein; the absorbent material comprising a hydrophobic and oleophilic material configured to absorb between two to eight times its weight of fats, oil, and/or grease; (b) forming a

biofuel by absorbing over a predetermined period of time fats, oil, and/or grease within the absorbent material such that the absorbed fats, oil, and/or grease within the absorbent material are separated from the water within the grease trap; (c) removing the biofuel from the grease trap while the separated water of step (b) remains in the grease trap; and (d) optionally transporting the biofuel to a secure location for storage and/or incineration. In certain aspects, step (d) is included in the above method. Within the above method, the absorbent material is loose, contained within a container comprising a tube adapted to receive and securely hold the absorbent material therein, and/or formed as a mat. Within the above method, the absorbent material consists of at least one of sphagnum peat, mushroom compost, orange peels, and polypropylene. Within the above method, the absorbent material is configured to absorb between four to eight times its weight of the fats, oil, and/or grease. Within the above method, before step (a) the volume of a grease trap and fats, oil, and/or grease relative to the water therein are visually assessed to determine an amount of absorbent material to be added to the grease trap. In certain aspects, the above method further includes after step (c), repeating steps (a)-(c) until a desired amount of fats, oils, and/or grease have been absorbed and removed from the grease trap or until all fats, oils, and/or grease have been absorbed and removed from the grease trap. In certain aspects, the above method further includes after step (b), repeating steps (a) and (b) until a desired amount of fats, oils, and/or grease has been absorbed by the absorbent material in the grease trap or until all fats, oils, and/or grease has been absorbed by the absorbent material. In certain aspects, the above method further includes pumping air by a positive pressure pump into the mixture of water, fats, oil, and grease to increase absorption, coagulation, and/or flocculation of the absorbent material during steps (a) and (b) and to aid and facilitate removing the biofuel from the grease trap during step (c) by maintaining the biofuel on an uppermost surface of the separated water that remains in the grease trap.

(46) According to another aspect of the invention, a vacuum truck for capturing and disposing of fats, oil and/or grease is disclosed, where the vacuum truck includes (1) a tank for housing fats, oil and/or grease, where the tank is also configured for housing loose absorbent material; (2) a vacuum hose coupled to the tank at an inlet of the tank, where the hose is configured for vacuuming the fats, oil and/or grease from a grease trap; and (3) an outlet of the tank, where the outlet is configured for opening to discharge any contents contained within the tank.

(47) According to another aspect of the invention, a method of capturing and disposing of fats, oil and/or grease includes (1) providing a vacuum truck to a user, where the truck includes a tank having a hose, an inlet, an outlet, and one or more mixing components; (2) positioning within the vacuum truck an absorbent material capable of absorbing a quantity of fats, oil and/or grease, where the absorbent material is selected from the group consisting of sphagnum peat, mushroom compost, and/or orange peels, and combinations thereof; (3) vacuuming fats, oil and/or grease from a grease trap via the vacuum hose, where the fats, oil and/or grease are vacuumed through the vacuum hose, through the tank inlet, and into the tank; (4) mixing the absorbent material and the vacuumed fats, oil and/or grease via the one or more mixing components within the tank to create a biofuel; and (5) discharging the biofuel from the tank into a containment device at a burn facility or other centralized location.

(48) According to another aspect of the invention, a method of capturing and disposing of fats, oil and/or grease includes the steps of: (1) providing a vacuum truck to a user, where the vacuum truck includes a tank having a hose, inlet, and outlet; (2) vacuuming fats, oil and/or grease from a grease trap via the vacuum hose, through the tank inlet, and into the tank; (3) discharging the fats, oil and/or grease from the tank into a containment device at a burn facility or other centralized location, where the containment device houses an absorbent material or an absorbent material is later added to the containment device, where the absorbent material is capable of absorbing fats, oil and/or grease, where the absorbent material is selected from the group consisting of sphagnum peat, mushroom compost, orange peels, and combinations thereof; and (4) mixing the absorbent

material and the vacuumed fats, oil and/or grease via one or more mixing components within the containment device to create a biofuel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of the geotextile container in tube form according to an embodiment of the invention;
- (2) FIG. 2 is a fragmentary, enlarged end view of the tube of FIG. 1, shown in an open position for receiving a quantity of sphagnum peat material or other absorbent material;
- (3) FIG. 3 is a perspective view of a geotextile container in mat form according to an embodiment of the invention;
- (4) FIG. 4 is a vertical cross-section of the geotextile mat with sphagnum peat material or other absorbent material contained in the mat;
- (5) FIG. 5 schematically illustrates the processes by which the biofuel is processed into various end use configurations;
- (6) FIG. 6 schematically depicts an alternative method of the biofuel formation/production and removal from, for example, a grease trap using the disclosed absorbent material(s);
- (7) FIG. 7 is a side view of an exemplar vehicle that may be used in collecting fats, oils, and grease from a grease trap;
- (8) FIG. 8 is a side view of the vehicle of FIG. 7 containing a quantity of sphagnum peat material or other absorbent material within a tank of the vehicle;
- (9) FIG. 9 is a side view of the vehicle of FIG. 7 containing a quantity of fats, oils, and grease ("FOG") within the tank of the vehicle;
- (10) FIG. 10 is a side view of the vehicle of FIG. 7 containing a quantity of sphagnum peat material or other absorbent material and a quantity of fats, oils, and grease ("FOG") within the tank of the vehicle, which combine to create the biofuel;
- (11) FIG. 11 is a side view of the vehicle of FIG. 7 illustrating an exemplary mixing element located within the tank of the vehicle;
- (12) FIG. 12 is a side view of the vehicle of FIG. 7 illustrating an exemplary heating element located within the tank of the vehicle;
- (13) FIG. 13 is a side view of the biofuel of FIG. 10 being deposited into a collection container for later use and/or burning of the biofuel;
- (14) FIG. 14 is a perspective view of an exemplary burn facility where the FOG and/or the FOG and peat or absorbent material may be discharged from the vehicle and subsequently burned.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(15) Referring now to the drawings, a container in the form of a tube **10** for use in the present invention is shown in the FIGS. 1-2, as noted. The tube **10** may be constructed according to many suitable constructions, but one construction comprises an elongate tube **10** that is formed of a geotextile fabric **12** that may be constructed by circular knitting, flat knitting, weaving, non-woven formation or any other fabric construction having a multitude of openings through the thickness of the fabric **12**. The fabric **12** is preferably seamed along its length or circular knitted to form the tube **10**. The tube **10** is preferably constructed of a biodegradable or natural material that will combust with minimal residue. The fabric **12** of the tube **10** may be constructed of any suitable natural or biodegradable yarn, for example, with a natural fiber such as cotton, hemp, ramie, jute or similar material because of its biodegradable characteristics, with apparent opening size (AOS) on the order of 0.25 to 0.5 mm depending on the size of the sphagnum peat absorbent material, mushroom compost absorbent material, orange peels as absorbent material, and/or the polypropylene absorbent material positioned therein (as the absorbent material). The empty tube **10**

may be any suitable length and diameter, for example, 60 cm to 120 cm long and 7 cm to 15 cm in diameter depending on the size of the grease trap and the FOG loading from the restaurant or vehicle repair facility. As manufactured, the tube **10** is preferably closed at one end and filled from the opposite, open end. The open end of the filled tube **10** may be closed with any suitable closure, such as stitching, clips or tied off with cord at the top of the grease trap or other FOG separating and collection structure.

(16) The tube **10** may include an opening **14** on either or both ends to receive a cord **16**, as shown in FIG. **1**, by which the tube **10** may be lowered into and retrieved from a grease trap or other enclosure and tethered to the grease trap or other structure while in use.

(17) One or more coatings may be applied to the fabric **12** to prevent penetration of the fabric **12** surface by water or aqueous salts thereby allowing the fabric **12** substrate to be non-absorbent for water or soluble salts.

(18) Referring now to FIGS. **3** and **4**, a container in the form of a mat **20** for use in the present invention is shown in the FIGS. **3-4**, as noted. The mat **20** may be constructed according to many suitable constructions, but one construction comprises a rectangular “box” shape that is formed of a geotextile fabric **22** that may be constructed by circular knitting, flat knitting, weaving, non-woven formation or any other fabric construction having a multitude of openings through the thickness of the fabric **22**. The fabric **22** is preferably seamed along its length and width to form the mat **20**. The mat **20** may be constructed of a synthetic, biodegradable or natural material. The fabric **22** of the mat **20** may be constructed of any suitable natural or biodegradable/synthetic yarn, for example, a natural fiber such as cotton, hemp, ramie, jute or similar material because of its biodegradable characteristics, with apparent opening size (AOS) on the order of 0.25 to 0.5 mm depending on the size of the sphagnum peat or mushroom compost absorbent material. The empty mat **20** may be any suitable length, width and height, for example, 60 cm to 120 cm long, 30 cm to 60 cm long and 10 cm to 20 cm in height depending on the size of the grease trap and the FOG loading from the restaurant or vehicle oils from the vehicle repair facility. As manufactured, the mat **20** is preferably closed at one end and filled from the opposite, open end. The open end of the filled mat **20** may be closed with any suitable closure, such as stitching, clips, or snaps **24**. The mat **20** may be seamed in such manner as to create individual compartments within the mat **20**.

(19) The mat **20** may include a loop **26** to receive a cord by which the mat **20** may be lowered into and retrieved from a grease trap or other enclosure and tethered to the grease trap or other structure while in use.

(20) One or more coatings may be applied to the fabric **22** to prevent penetration of the fabric **22** surface by water or aqueous salts thereby allowing the fabric **22** substrate to be non-absorbent for water or soluble salts.

(21) The preferable FOG absorbent material filled into the mat **20** is a specialized form of sphagnum peat “SP”, mushroom compost materials, and/or orange peels. In alternative aspects, the absorbent mater is a polymeric material having oleophilic and hydrophobic properties that are ideal to absorb fat, oils, and/or grease. In this aspect, the absorbent porous, polymeric material is a polyolefinic material and more preferably a polypropylene having oleophilic and hydrophobic properties that are ideal to absorb fat, oil, and/or grease.

(22) The following step by step process is expected for typical use and implementation of the FOG product and collection process. The process is explained with reference to the tube **10**, but will be essentially the same when using the mat **20**.

(23) STEP 1: Introduce the tube **10** with selected sphagnum peat “SP” or mushroom compost into the grease trap or other FOG collection structure. Tether the tube **10** with cord **16** so that it stays at the influent end of the grease trap, and is the optimal location of FOG collection. Prior to placement in the grease trap, weigh the dry tube **10** so that a ‘before and after” measure of FOG collection can be established.

(24) STEP 2: After consultation with local water and sewer regulatory officials and the owner of

the FOG collection device or grease trap, setup of a regular interval to remove and replace the FOG collection absorbent tube **10**. From past experience, the best way to initiate the use of the FOG remediation technology is to start off as a regulatory approved Demonstration Project where the approach and results are measured and evaluated.

(25) STEP 3: Depending on the interval for removal and collection of the FOG absorbent tube **10**, arrange for storage in covered and secured FOG containers to avoid attracting small animals and rodents that are common in and around restaurants and vehicle repair facilities.

(26) STEP 4: Transport the FOG tubes to an SSI facility. The FOG tube **10** is then part of waste to energy, renewable energy biofuel source.

(27) The advantage of the FOG process using the tube **10** or mat **20** is that it safely and cost effectively separates FOG in the grease trap before it is mixed with large volumes of water and emulsified waste liquids. Separation after the fact is difficult and expensive.

(28) The FOG absorbent tube **10** works for FOG collection because the sphagnum peat “SP”, mushroom compost material, orange peel, and/or polypropylene are highly absorbent natural or polymeric materials that separate the FOG from liquids or water. In other words, each of the absorbent material has both oleophilic and hydrophobic properties. The absorbing characteristics advantageously include combination of increased surface area and natural filtering processes, similar to that provided by charcoal or activated carbon. A slightly larger AOS in the filtering geotextile fabric **12** will allow more of the natural absorbing and geochemical attraction between the sphagnum peat “SP” to have better contact with the surface FOG materials to attract and collect it from the liquids/water. This approach reduces the tendency or emulsification of the FOG into the grease trap so that frequency of the grease trap pumping and remixing of the FOG and water/liquid will be reduced. Collecting the FOG from the surface of the grease trap is much more efficient and cost effective.

(29) Estimate of the absorbing qualities of peat moss appear to be in the range of 5 to 10 kg/m² per FOG tube **10** per week. This will be an area of applied research and measurement during future demonstration projects.

(30) Polar molecules have a positive charge on one end and a negative charge on the other end. Non-polar molecules do not have two electrical poles and the electrons are distributed symmetrically on both sides. FOG is composed of organic non-polar compounds. Water is a polar solvent. Only polar compounds or other polar solvents will mix with water. Therefore, non-polar FOG will not readily mix with water. Depending on the source, FOG has a density of approximately 0.863-0.926 g/cm³. Water has a density of approximately 1.000 g/cm³. The lesser density substance will float on top of the greater density substance if it does not mix, thus non-polar FOG floats on water because it does not mix and gravity exerts more pull on the greater density water molecules. Water molecules are relatively small because they are only composed of one oxygen and two hydrogen molecules (H₂O). They, therefore, pack closely together in a space. Molecules of oil are large and have complicated shapes, thus requiring more space than water molecules, which is why oil is less dense than water.

(31) A few oils having densities less than water are known to be polar compounds and can mix with water and therefore not float on the water's surface.

(32) Thus, polarity and density both contribute to oil floating on water.

(33) Polarity is a relative term. On a sliding scale, some oils are more or less polar than others, and have both polar and non-polar attributions. Also, the heating of oils and interaction with other organic compounds it is exposed to during heating, can change the oil's chemical composition, and thus change the relative polarity.

(34) The above referenced principles permit the method of this application to work as intended and as developed.

(35) Further evidence supporting the “charge” principle is found at Fat, Oil and Grease Science, Dothan, Alabama. Fat, Oils, and Grease (FOG) Science.

(36) Testing of FOG as described above returned a B.T.U. value of 14,019 per pound using a method identified as ASTM D240. According to a preferred embodiment of the invention, the FOG product contains between 88-75 percent FOG and between 12 and 25 percent peat or mushroom solids as described above. Expected range of B.T.U. output is 12,500 to 15,500 B.T.U. per pound, preferably approximately 14,000 B.T.U. The biofuel can be transported in its original container and subsequently compressed into a pellet, or log or other shape, shredded or granulated to increase its surface area and render it more easily combustible. In an alternate embodiment, as described further below, the biofuel may be transported by a vacuum truck or other suitable vehicle and subsequently burned at a burning facility.

(37) Motor vehicle oils similarly incorporated into the FOG product can produce in the range of 20,000 B.T.U. per pound.

(38) If the FOG is originally collected in a synthetic container, transferring the FOG into some form of container of natural materials, as described above, means that the entire product, FOG, capture material, and container can be used as fuel. As is apparent from the above discussion, the FOG/capture material product can be removed from its formation container for being compressed into a pellet, log, cake or other shape, shredded or granulated, or may remain in its formation container for being combusted, as illustrated in FIG. 5. In some embodiments, the FOG is not transferred into a container of capture materials, but rather captured or collected using a vacuum truck or other suitable vehicle having either a capture material already within the tank of the truck, or at the burning facility where the FOG will be transferred to to be used as fuel once mixed with one or more mixing components with the absorbent mater or polymeric material having oleophilic and hydrophobic properties that are ideal to absorb fat, oils, and/or grease.

(39) A biofuel product having fat, oil and/or grease components according to the invention has been described with reference to specific embodiments and examples. Various details of the invention may be changed without departing from the scope of the invention.

(40) As further shown in FIG. 6 and in additional aspects, the FOG absorbent material may be used alone (i.e., excluding the previously mentioned container that contains the absorbent material and/or the absorbent material separated from the container) to absorb FOG when forming/producing the biofuel product (i.e., the absorbent material having FOG absorbed therein). When used in this manner, a user (e.g., technician) places a desired, predetermined amount of absorbent material within, for example, a grease trap (or other container having FOG therein). The user then allows an adequate time for the absorbent material to absorb and preferably become saturated with FOG thereby forming the biofuel product. Subsequently the user removes the biofuel product from, for example, the grease trap for subsequent use and/or processing steps to be used as the biofuel product. During the removal step of the biofuel product (i.e., absorbent material having FOG absorbed therein) in the above mentioned process, it is further envisioned that a positive pressure pump (e.g., a modified pool pump such as the Pentair line of pool pumps) and other removal aids (e.g., a net, filter, vacuum truck, etc.) will be used to further expedite the overall process to remove the biofuel from the grease trap in an expeditious and efficient manner.

(41) More specifically in the above-mentioned method and in view of FIG. 6, the technician arrives at the grease trap to be serviced and removes/opens the grease trap's lid (S1). Next, the amount of absorbent material to be introduced into the grease trap will be determined by the technician visually inspecting and assessing the overall volume of the grease trap along with the ratio of FOG to brown water contained therein (S2). The technician next introduces a sufficient amount of the loose absorbent material (e.g., 1 pound, 5 pounds, 10 pounds, 20 pounds, etc.—termed the “first amount” or “initial amount”) into the trap (S3) to allow for the desirable amount of absorption of FOG into the absorbent material. For example, each pound of absorbent material should absorb between 5 pounds to 9 pounds of FOG per pound of absorbent material, and in preferred aspects, each pound of absorbent material should absorb between 7 pounds to 8 pounds of FOG per pound of absorbent material. In preferable aspects, the desirable amount of FOG absorption of FOG into

the absorbent material is 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 98%, or 100% and any percentage there between of the FOG within the grease trap, and in certain aspects the percentage removal of FOG complies with local laws and ordinances, state laws and ordinances as well as any applicable Federal laws.

(42) After a predetermined time period in which absorption of FOG into the absorbent material has occurred (S4), the grease trap is again visually inspected and assessed to further determine whether any FOG remains (S4'), and if so, to further determine the ratio of FOG to brown water remaining in the grease trap. At that time, more absorbent material (a "second amount" of absorbent material) may be added (S5.sup.a') to absorb the remaining FOG (i.e., FOG that was not absorbed by the previously provided absorbent material in (S3)). After a predetermined time period (S5.sup.a''), the technician again visually assesses whether any FOG remains in the grease trap (outside of the biofuel) or whether the desired amount of FOG has not been absorbed. If the answer to (S5.sup.a'') is no, the biofuel is removed from the grease trap (S6) by the removal process discussed further below and subsequently stored (S7).

(43) However, if the answer to (S5.sup.a'') is yes, subsequent amounts of absorbent material (e.g., "third amount", "fourth amount", "fifth amount", etc.) may be added (S5.sup.a''), by the technician until the desirable amount of FOG absorption (and desired FOG to brown water ratio) has been reached. After reaching the desired FOG absorption amount in the absorbent material (and the desired FOG to brown water ratio), the absorbent material having FOG absorbed therein (biofuel) may be subsequently removed (S6) all at once by the removal step/process discussed further below.

(44) As an alternative to the above, an absorbing step followed by an immediate removal step may be used. In this aspect, the absorbent material (S3) (i.e., first amount or initial amount) is added to the grease trap and after a predetermined time period (S4) FOG is absorbed into the absorbent material forming the biofuel. Alternatively, a vehicle containing absorbent material may remove the FOG from the grease trap and form the biofuel within a tank of the vehicle by combining the FOG with the absorbent material within the tank. If no FOG remains outside of the absorbent material or the desired amount of FOG has been absorbed thereby forming the biofuel (S5), the biofuel (S6) (i.e., first amount of absorbent material having FOG absorbed therein) is removed from the grease trap and the process is concluded by subsequently storing (S7) the biofuel.

(45) However, in certain alternative aspects and after steps (S4) and (S4'), it may be determined that the absorbent material has absorbed FOG but some FOG remains outside of the absorbent material or the desired amount of FOG was not absorbed (S5.sup.b) by the absorbent material. (S6) removal of the absorbent material having FOG absorbed therein (i.e., biofuel) from the grease trap may immediately occur after (S5.sup.b), and the technician may again visually inspect and assess the grease trap to determine whether any additional FOG remains therein. If so, the technician may again, introduce (S5.sup.b) the absorbent material ("second amount" of absorbent material) and allow for absorption of FOG into the second amount of absorbent material (biofuel) (S5.sup.b''). If no FOG remains outside of absorbent material or desired amount of FOG has been absorbed, the biofuel is removed from the grease trap (S6). However, after (S5.sup.b'') if FOG remains outside of absorbent material or desired amount of FOG has not been absorbed, (S5.sup.b'') occurs in which (S6), (S5.sup.b), (S5.sup.b''), and (S6) are repeated until no FOG remains outside of the absorbent material or the desired amount of FOG has been absorbed (e.g., almost all FOG has been absorbed), the biofuel from the grease trap and subsequently stored (S7).

(46) To further aid in the removing/removal steps (i.e., removing the biofuel comprising the absorbent material having FOG absorbed therein), additional removal tools and aids may be used. For example, the technician may use a rake or rake-like device to actuate the absorbent material, FOG, and/or brown water in the grease trap and to spread the absorbent material more evenly on the surface of the FOG/water mixture within the grease trap. Alternatively, the technician may use a hose from a vacuum truck to collect the FOG. This step advantageously increases the rate of absorption into the absorbent material and further expedites the FOG capture and removal process

from the grease trap. During this step, the absorbent material having FOG absorbed therein is hydrophobic and will continue to float even when saturated with FOG.

(47) At this point, the technician may further utilize a positive pressure pump (a modified pool pump such as the Pentair line of pool pumps, or a vacuum truck) to pump the biofuel (i.e., FOG laden absorbent material) from the top, or uppermost, portion of the grease trap and to remove the biofuel therefrom via a filter and/or physically remove (e.g., via a net or another similar removal device). During the removal process and while using the pump or vacuum truck, the pump/vacuum truck will beneficially circulate the brown water back into the grease trap thereby keeping the grease trap maximally efficient and effective. The maximum efficiency of the grease trap is achieved by keeping the grease trap full of water during the removal process.

(48) While pumping the FOG laden absorbent material out of the grease trap, the technician moves the end of the hose around the top and/or uppermost sections of the grease trap surface until all of the biofuel (i.e., absorbent material having FOG absorbed therein) has been removed from the grease trap.

(49) After the biofuel (i.e., absorbent material laden with FOG and/or absorbent material having FOG absorbed therein) has been captured and removed from the grease trap and the unencumbered water drained from the positive pressure pump hoses into the grease trap, the grease trap lid is closed securely. Then the technician either transports the biofuel to another location or leaves the biofuel in a secure area for subsequent removal and processing.

(50) In still further alternative embodiments, and in view of FIG. 7, the technician may open the grease trap's lid, but rather than introducing the loose absorbent material, the technician inserts a hose **102** or other vacuum component of a vacuum truck **100** into the grease trap to capture and collect the FOG to transport it to a burn facility or other centralized location **122**. In particular embodiments, the vacuum truck **100** or mobile FOG transportation vehicle includes a driver/passenger cabin **104**, a tank **106** for containing the FOG, one or more wheels **108**, a vacuum hose or other inlet device **102**, and an outlet **110** for discharging or removing the FOG from the vehicle. The vacuum truck includes, as may be known and/or understood in the art, the following features generally including but not limited to safety features, a debris tank body, a vacuum system, a water system, a boom system, a tailgate system, a boom tube, a receiver tank as stated, a separator, a filter, a suction source, one or more safety valves, at times an integrate vapor recovery system, one or more additional intakes with internal pipes, a grounding reel, in some embodiments—an air cooled pump, one or more adjustable hose hooks, one or more hose reels, one or more discharge valves, a tank level indicator, sight eyes, vacuum/pressure gauge, manway/hatch/portal, vacuum relief valve, pressure relief valve, primary shutoff, secondary shutoff, float ball, vacuum pump, oil catch muffler/oil separator, precision controls, tank rails, grit ridder, blower/moisture trap and grit ridder, moisture trap, fresh water tank, storage or hybrid attachments/wands/hoses, manual and hydraulic door locks and lifts, standard chassis, vacuum blower, among other components.

(51) In some embodiments, the mobile FOG transportation vehicle **100** may further include one or more heating elements **112**, as shown in FIG. 12. Additionally, the vacuum truck **100** may be filled with a proportionate amount of loose peat or another absorbent material **114**, as detailed above, prior to a technician utilizing the vehicle to extract the FOG from a grease trap. In alternate embodiments, as shown in FIG. 9, the vacuum truck may simply be used to collect the FOG **116** and transport it to the burn facility **122** where it can be mixed with peat or another absorbent material **114** to create the biofuel **118**. Similar to determining the amount of loose peat or other absorbent material **114** that may fill the tank **106**, the vacuum truck **100** may also be limited on its capacity for carrying FOG **116** and/or the proportion of FOG collected may be determined by how much peat it will be mixed with at the burn facility **122**.

(52) According to example embodiments, the vacuum truck **100**, as shown in FIG. 8, includes a preferred amount of peat or other absorbent material within the recovery tank **106** of the vehicle. The amount of loose peat or other absorbent material **114** may be determined, similar to above, by

calculating the specific vehicle's tank **106** size based on capacity and carrying weight. In some embodiments, the amount of loose peat or other absorbent material **114** may account for approximately 1/5-4/5 of the available space in the tank, more particularly, between about 2/5-4/5, for example about 4/5, or as predetermined to allow full use of the space with absorbent material. In other embodiments, the amount of loose peat or other absorbent material **114** will be proportionate to the amount of FOG **116** that needs to be vacuumed as well as the capabilities (or limitations) of the particular vacuum truck **100** being used. Upon vacuuming the FOG **116**, as shown in FIG. **10**, from the grease trap, the absorbent material **114** and the FOG **116** are combined to create the biofuel **118** (i.e., absorbent material laden with FOG and/or absorbent material having FOG absorbed therein). The FOG **116** and absorbent material **114** may be combined by various mixing elements or components **120** (see FIG. **11**) that stir and/or move the FOG until it is generally fully absorbed by the loose peat or other suitable absorbent material.

(53) Referring again to FIG. **12**, in various embodiments, the mobile FOG containment vehicle **100** may include a heating element **112** within or directly coupled to the tank **106** of the vehicle in order to prevent the FOG **116** and/or FOG biofuel **118** from congealing while it is transported to a burn or other processing facility **122**, as shown in FIG. **14**, or from congealing while capillary absorption process ensues to create the mixing process. In particular embodiments, the heating elements **112** keep the FOG **116** from congealing by keeping the temperature between about 100°-250°, or more preferably about 110°-200°, for example about 130°-150°. After transporting the FOG **114** or FOG fuel **118** to a burn facility **122**, for example, the FOG/FOG fuel is removed or discharged from the vehicle via the outlet **110** and subsequently burned. Referring to FIG. **13**, in some embodiments, the FOG fuel **118** or similarly the FOG may be discharged by the tank such that the FOG fuel may be released from the tank of the vehicle into a capture device **124** at the particular facility **122**. In other embodiments, a drain or other hose **124** may be used to allow the FOG fuel **118** to be drained from the vehicle.

(54) Furthermore, the foregoing description of the preferred embodiments of the invention and best mode for practicing the invention are provided for the purpose of illustration only and not for the purpose of limitation of the invention being defined by the claims.

Claims

1. A vacuum truck for capturing and disposing of fats, oil and/or grease, the vacuum truck comprising: (a) a tank for housing fats, oil and/or grease, wherein the tank is also configured for housing loose absorbent material; (b) a vacuum hose coupled to the tank at an inlet of the tank, wherein the vacuum hose is configured for vacuuming the fats, oil and/or grease from a grease trap; (c) an outlet of the tank, wherein in the outlet is configured for opening to discharge any contents contained within the tank; (d) a driver/passenger cabin located at the front end of the vacuum truck; and (e) one or more wheels supporting the tank and the driver/passenger cabin.
2. The vacuum truck according to claim 1, wherein the tank comprises one or more mixing elements for stirring any contents contained within the tank.
3. The vacuum truck according to claim 2, wherein the contents contained within the tank comprise a combination of fats, oil and/or grease, and a loose absorbent material.
4. The vacuum truck according to claim 3, wherein the mixing elements stir the contents contained within the tank to create a biofuel.
5. The vacuum truck according to claim 4, wherein when the outlet is opened, any contents contained within the tank are discharged into a containment device at a burn facility.
6. The vacuum truck according to claim 5, wherein the vacuum truck further comprises a tilting feature to assist with discharging any contents contained within the tank into the containment device at the burn facility.
7. The vacuum truck according to claim 2, wherein the contents contained within the tank comprise

a combination of fats, oil and/or grease collected from a grease trap using the vacuum hose coupled to the tank inlet.

8. The vacuum truck according to claim 7, wherein the contents contained within the tank are discharged into a containment device containing an absorbent material at a centralized location.

9. The vacuum truck according to claim 7, wherein the tank comprises one or more heating elements for warming any contents contained within the tank to prevent the contents from congealing.

10. A method of capturing and disposing of fats, oil and/or grease, the method comprising the steps of: (a) providing the vacuum truck according to claim 1 to a user, wherein the vacuum truck further comprises one or more mixing components; (b) positioning within the vacuum truck tank an absorbent material capable of absorbing a quantity of fats, oil and grease, wherein absorbent material is selected from the group consisting of sphagnum peat, mushroom compost, orange peels, and combinations thereof; (c) vacuuming fats, oil and/or grease from a grease trap via the vacuum hose, wherein the fats, oil and/or grease are vacuumed through the vacuum hose, through the tank inlet, and into the tank; (d) mixing the absorbent material and the vacuumed fats, oil and/or grease via the one or more mixing components within the tank to create a biofuel; and (e) discharging the biofuel from the tank into a containment device at a burn facility.

11. The method of claim 10, further comprising: (f) burning the biofuel at the burn facility.

12. The method of claim 10, wherein the tank further comprises one or more heating elements for warming the absorbent material and vacuumed fats, oil and/or grease within the tank to prevent the absorbent material and the fats, oil and/or grease from congealing.

13. The method of claim 12, wherein the one or more heating elements keep the absorbent material and the fats, oil and/or grease at a temperature between approximately 110 degrees and 200 degrees.

14. The method of claim 12, wherein the one or more heating elements keep the absorbent material and the fats, oil and/or grease at approximately 140 degrees.

15. A method of capturing and disposing of fats, oil and/or grease, the method comprising the steps of: (a) providing the vacuum truck according to claim 1 to a user; (b) vacuuming fats, oil and/or grease from a grease trap via the vacuum hose, wherein the fats, oil and/or grease are vacuumed through the vacuum hose, through the tank inlet, and into the tank; (c) discharging the fats, oil and/or grease from the tank into a containment device at a burn facility, wherein the containment device houses an absorbent material capable of absorbing the fats, oil and grease, wherein absorbent material is selected from the group consisting of sphagnum peat, mushroom compost, orange peels, and combinations thereof; and (d) mixing the absorbent material and the vacuumed fats, oil and/or grease via one or more mixing components within the containment device to create a biofuel.

16. The method of claim 15, further comprising: (e) burning the biofuel at the burn facility.

17. The method of claim 15, wherein the tank further comprises one or more heating elements for warming the vacuumed fats, oil and/or grease within the tank to prevent the fats, oil and/or grease from congealing.

18. The method of claim 17, wherein the one or more heating elements keep the fats, oil and/or grease at a temperature between approximately 110 degrees and 200 degrees.

19. The method of claim 17, wherein the one or more heating elements keep the fats, oil and/or grease at approximately 140 degrees.

20. The method of claim 15, wherein the tank further comprises one or more mixing components.

21. The vacuum truck according to claim 1, wherein the vacuum truck further comprises the loose absorbent material within the tank.
